METRIC

A-A-52442 (AT)
March 24, 1993
SUPERSEDING
MIL-R-45949A(AT)
23 May 1977

COMMERCIAL ITEM DESCRIPTION

REPAIR KIT, PUNCTURE, PNEUMATIC TIRE (METRIC)

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-R-45949A(AT) which is canceled.

1.0 Abstract. This commercial item description (CID) covers. one type of tubeless tire repair kit which will allow emergency "outside-in" repair of all types and sizes of tubeless heavy-duty tires up to and including 28-ply or ply-rated construction. The kit is to be used to repair tires while mounted on the vehicle.

2.0 Salient characteristics.

2.1 Description. The tire repair kit covered by this CID is for field use on tactical vehicles. The kit shall allow a temporary “get-home” capability of 65 kilometers minimum.

2.2 Material. All parts and materials used in the fabrication of the repair kit shall be of a quality necessary to produce a kit which shall meet the requirements specified herein. The use of recovered material made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see 5.4).

2.2.1 Hazardous material. The items that comprise the tire repair kit shall not contain any hazardous material as defined in FED-STD-313 which may be toxic or harmful to personnel when used for its intended purpose. These items must also be considered non-hazardous in accordance with national and International regulations for the transportation and storage of dangerous goods.

Beneficial comments recommendations, additions, deletions clarification, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 2640

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2.3 Design and construction. The repair kit shall be designed and constructed so that it can be used for the operations specified herein. Unless otherwise specified (see 5.2), the repair kit shall consist of the following components and in the quantities (Qty) indicated:

Component	Qty
Insertion tool, handle type	1 each
Repair material	1525 centimeters (cm) total length
Bonding compound	470 milliliters (ml) minimum
Leak detection material	60 ml minimum
Leak detection applicator bottle	1 each
Skiving knife	1 each
Reamer tool	1 each
Hinged box	1 each

2.3.1 Details of components.

2.3.1.1 Insertion tool. The insertion tool shall consist of a corrosion resistant steel insertion needle with either a permanently attached or detachable handle. The handle shall be a heavy duty (truck) type of a suitable size and configuration to allow the insertion needle to perform its intended function without injury or stress to the operator's hand. The needle shall be of sufficient diameter, strength, and hardness as to be suitable for its intended purpose. It shall have a suitable eye to accommodate the repair material to be used for insertion into injury of tubeless tires. The needle, whether adjustable or permanently fixed in length, shall be suitable in length to perform its intended function. It shall be capable of withstanding a longitudinal force of 450 newtons without breaking or being permanently deformed.

2.3.1.2 Repair material. The repair material shall be polypropylene or nylon sleeving impregnated and coated with a polysulfide or butyl rubber compound and have a minimum tensile strength of 35,160 kilopascals (kPa). The repair material shall be round or elliptical in shape. It shall be furnished with cross-sectional diameter adequate to seal punctures ranging from nail holes Up to holes 19 mm in diameter and to seal slits 127 mm in length. A minimum total length of 1525 cm of repair material shall be furnished in the kit. The repair material shall be coiled and encapsulated in plastic tubes or cut to appropriate lengths and packaged five to ten individual refills per packet. Provisions for capping ends of coiled repair material tubes shall be provided to retain repair material in tube when not in use. Cut-to-length repair material shall be protected by a backing of "release" parchment paper on one side and a mylar covering on the opposite side. The packets shall be sealed in polyethylene bags appropriate to the refill size. The tubes or bags for the repair material shall be properly identified as to their contents and use (see 2.6).

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2.3.1.3 Bonding compound. The bonding compound shall be a fast self-drying type suitable to perform its intended function A minimum of 470 ml shall be furnished in each kit. The material shall be in at least two screw-top cans or plastic jars. The containers shall be properly identified as to content and have reasonable unlimited storage life under normal conditions (see 2.6).

2.3.1.4 Leak detector. The leak detector material shall be in the form of water soluble concentrated liquid. When properly mixed with water, it shall have the capacity of detecting and showing leakage or seepage of pressure from a tubeless tire. A minimum of 60 milliliters (ml) of leak detector material shall be furnished in each kit. It shall be contained in a screw-cap type plastic container which shall be leak-proof and capable of preventing moisture absorption from the air. It shall under normal conditions have a reasonably unlimited storage life and be identified as to its contents and capacity (see 2.6).

2.3.1.5 Leak detector application bottle. The leak detector bottle shall be of plastic, of at least 120 ml capacity, squeeze dispensing-type for mixing the leak detector material with water before use and for application of mixed solution to tubeless tire to allow visual detection of any leak or seepage of pressure. The applicator bottle shall incorporate a screw-type applicator cap for filling and dispensing and an attached cap for applicator tip. It shall also be properly identified as to its use (see 2.6).

2.3.1.6 Skiving knife. A skiving knife with a blade length suitable to perform its intended function shall be provided. The blade shall have a square point and be of high quality, sharpenable steel permanently mounted in a handle.

2.3.1.7 Reamer tool. The reamer tool shall consist of a round tapered reamer needle type instrument permanently mounted in a screwdriver type handle. The handle shall be heavy duty (truck) type of a suitable size to allow the reamer to perform its intended function without injury or stress to the operator’s hand. The reamer portion of the tool shall be hardened steel. It shall have a tapering diameter starting with a point and tapering out to a suitable diameter and length for its intended function. The latter two thirds of the reamer measured from the handle shall be serrated. Tool must be capable of being forced into, through, and enlarging the average nail type puncture at the thickest or stiffest portion of all types and sizes of tubeless tire casings up to and including 28-ply or ply-rated construction. Tool shall be capable of performing the described intended function without dislodging handle, bending, breaking or permanently deforming reamer portion of tool in any way.

2.3.1.8 Repair kit box. The repair kit box shall be fabricated of not less than 0.60 mm thick metal or equivalent strength plastic material and be rectangular in shape with all sides flat including the lid. The lid shall be hinged. The box and lid shall have a positive latching device. It shall be capable of containing all component parts of the kit and shall be painted in accordance with 2.5. See 2.6 for marking requirements.

2.3.1.9 Repair kit instructions. The repair kit shall include a set of instructions detailing procedures for detecting leaks and repairing tires using the components contained within. Instructions shall include a section on the safe and proper use of and disposal of the tire repair kit items (see 2.4). The

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instructions shall be printed on a plate or Label permanently affixed to the inside of the lid (see 2.6). The markings or printed information snail be permanent and impervious to solvents, cleaning fluids, or other repair kit chemicals.

2.4 Material. Safety data sheet. A MSDS shall be prepared in accordance with FED-STD-313 (see 5.2 and 5.5).

2.5 Finish treatment. Cleaning, pretreating, priming and painting of the repair kit box shall conform to an appropriate CARC paint system selected from Drawing No. 12369000 for internal application. Unless otherwise specified (see 5.2), color of paint final top coat shall be green 383.

2.6 Identification markings. Identification markings shall be placed on the outside of the repair box kit and shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE), the contract number, the part identification number (PIN) A52442-1, and the national stock number (NSN). All kit component markings and labels shall be legible and durable.

2.7 Workmanship. The repair kit shall be processed in such a manner to be free from cracked or misplaced parts, sharp edges, burrs and other defects which will adversely affect its serviceability or appearance. The repair kit shall be complete, new and unused.

3.0 Quality assurance provisions.

3.1 Responsibility for Inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order (see 5.2).

5.0 Notes.

(This section contains reformation of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government specifications and standards. FED-STD-313 "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities" is available from the Navy Publications and Printing Service Office, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

5.1.2 Other Government documents, drawings, and publications. Army drawing 12369000 "Chemical Agent Resistant Coating CARC Paint System Index" is available from the Contracting Officer, U.S. Army Tank-Automotive Command, Warren, MI 48397-5000.

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5.2 Ordering data. Acquisition documents must specify the following:

a. Title, nunmer and date of this commercial item description.
b. PIN number and quantity required.
c. Issue of the DODISS to be cited in the solicitation and the specific issue of individual documents referenced (see 5.1).
d. Kit components, if other than specified (see 2.3).
e. Identify activities requiring copies of completed MSDS and specify when the MSDS will be inspected (see 2.4 and 5.5).
f. Final top coat color of paint. if other than specified (see 2.5).
g. Selection of applicable level and packaging requirements (see 4.0).

5.3 Cross reference. Repair kits conforming to this CID are interchangeable/substitutable with repair kits conforming to MIL-R-45949A(AT).

5.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

5.5 MSDS. The contracting officer should identify those activities requiring copies of the completed MSDS prepared in accordance with FED-STD-313.

MILITARY INTERESTS:

Custodian
Army - AT

Review Activities
Army - SM
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 2640-A198)

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